

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

Annotated data is scarce in specialized domains such as medical imaging, which makes robust vision models hard to train. Self-supervised masked image modeling (MIM) is a promising remedy, but two gaps limit it: existing approaches largely rely on random high-ratio masking that is inefficient and poorly aligned with task semantics, and the region-aware variants that try to fix this depend on reconstruction heuristics or supervised signals that do not transfer across tasks and modalities. We close both gaps with *Mask What Matters*, and this paper makes three contributions. **First**, we introduce the first controllable text-guided masking framework for self-supervised medical image analysis: it uses vision-language models for prompt-based region localization and then applies differentiated masking that emphasizes diagnostically relevant regions while suppressing redundant background. **Second**, this controllable, annotation-free design delivers better semantic alignment, stronger representation learning, and cross-task generalizability within a single unified protocol. **Third**, across three imaging modalities—brain MRI, chest CT, and lung X-ray—Mask What Matters consistently outperforms existing MIM methods (e.g., SparK), with gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection, and it does so at substantially lower masking ratios (e.g., 40% vs. 70%). Together, these results establish that controllable, text-driven masking enables semantically aligned self-supervised learning and advances robust vision models for medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection are fundamental computer vision tasks, and deep learning has become the core technology driving their recent advances. Yet high-performing deep models typically depend on large-scale annotated datasets. In specialized domains such as medical image processing, high-quality annotations are scarce and expensive to obtain, and this bottleneck is precisely the barrier our work sets out to lower: it stands in the way of adopting deep learning where it is needed most.

Self-supervised learning (SSL) addresses this bottleneck by pretraining on unlabeled data, letting models learn effective representations without manual annotations [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Among SSL paradigms, MIM has become especially popular: by masking parts of an image and requiring the model to reconstruct them, MIM forces the model to capture structural and semantic patterns from visible context. It performs strongly across many downstream tasks and has been extended to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

The problem we target is that prevailing MIM approaches—such as masked autoencoders (MAE) [He et al. (2022)] and SparK [Tian et al. (2023)]—mask randomly, without alignment to the regions that matter downstream. This task-agnostic masking creates two limitations that motivate our contributions. (1) **Semantic misalignment**: the learned representations may not match downstream objectives, which reduces transfer performance. (2) **Wasteful high masking ratios**: to raise the odds of masking informative areas, existing methods resort to extremely high ratios (e.g., 75% in MAE), in-

flating reconstruction difficulty and the demand for data and compute. By contrast, natural language is compact and semantically dense, so it can be learned effectively even at low masking ratios [Devlin et al., 2019]. The reliance on high ratios in image-based MIM stems from information redundancy and the absence of semantic-aware masking, which together prevent models from focusing on truly critical regions.

These two limitations are sharpest in medical imaging, which is exactly why we study it here. Medical images are semantically sparse: diagnostically relevant information is confined to small localized regions (e.g., lesions or organs), while most of the image is redundant background. Medical imaging also spans diverse modalities—MRI, CT, and X-ray—with wide variation in the shape, size, and spatial distribution of task-relevant areas. These properties place stringent demands on both the efficiency and the generalizability of any masking strategy, and they define the bar our method must clear.

To meet that bar, we introduce a controllable, text-guided masking framework—**Mask-What-Matters (MWM)**. Given a user-specified, open-vocabulary description or phrase, a pretrained vision-language model supplies zero-shot localization cues that we convert into robust regions of interest (ROIs). MWM then applies region-specific, prompt-conditioned masking—higher ratios on semantically important areas (e.g., lesions/organs) and lower ratios on background—thereby injecting downstream semantics directly into the pretraining signal, without per-image reports or labels. This directly answers limitation (1) by aligning masking with task semantics and limitation (2) by making low masking ratios effective.

We summarize the paper’s three contributions, which we return to throughout the paper and revisit in the conclusion:

- **Contribution 1 — The first controllable text-guided masking framework for medical imaging.** MWM integrates a vision-language model (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and apply differentiated ROI-vs.-background masking, directly overcoming the semantic misalignment of random masking.
- **Contribution 2 — An annotation-free, backbone-agnostic design.** MWM needs no per-image reports or labels and slots into both ViT- and ConvNet-based MIM pipelines under a single unified protocol, making the approach broadly reusable.
- **Contribution 3 — Consistent gains across modalities and tasks.** On brain MRI, chest CT, and X-ray, MWM surpasses state-of-the-art MIM baselines (e.g., SparK) in classification, detection, and segmentation—while using far lower masking ratios—demonstrating the promise of text-driven generative pretraining in medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling began in natural language processing with BERT [Devlin et al., 2019], which learns contextual representations by masking and predicting tokens during pretraining. Building on this idea, [He et al., 2022] brought MIM to computer vision: image portions are masked during pretraining and the model reconstructs the missing regions. The paradigm generalizes strongly in both natural and medical image domains [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative methods include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian et al., 2023]: MAE uses high-ratio random masking to push ViTs toward high-level semantics; MedMAE adapts this to medical images with domain-specific tweaks; and SparK ports the strategy to convolutional backbones, exploiting their multi-scale features to strengthen pretraining.

The gap our first contribution addresses is that these conventional MIM approaches mask randomly or structure-agnostically, ignoring the semantic requirements of downstream tasks. Such non-selective masking can lead the model to overemphasize irrelevant regions during pretraining, hurting transfer. Recent region-aware masking strategies try to improve semantic alignment, and they fall into two lines—each of which leaves a limitation that MWM is designed to remove.

The first line uses loss-driven adaptive masking: methods such as hard patches mining (HPM) [Wang et al., (2023a)] and its extension AnatoMask [Li et al., (2024)] track reconstruction error across regions and mask high-loss areas more aggressively, hoping to prioritize semantically valuable structures. The limitation is that high loss need not mean high task relevance: the model can be steered toward hard-to-reconstruct but task-irrelevant regions. In brain MRI, for instance, complex structures like the spinal cord or eyeballs may draw attention even though they are unrelated to downstream tumor detection.

The second line adds external perceptual modules to guide masking. FocusMAE [Basu et al., (2024)], for example, uses pretrained Region Proposal Networks (RPNs) to pick high-information regions for selective masking. The limitation here is supervision cost: the RPNs must be trained on domain-specific annotated data, which caps cross-task generalization.

Together these limitations—supervision cost, misalignment between masked regions and task semantics, and rigidity—motivate the more flexible, semantically aligned, and generalizable masking mechanism we propose. Natural language, a rich and interpretable source of supervision, is a promising way to guide masked modeling in a task-aware yet annotation-free manner, and is the foundation of our approach.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Rapid progress in vision-language alignment now makes it feasible to perceive and identify task-relevant regions under zero-shot and open-world settings [Radford et al., (2021)]. The Contrastive Language-Image Pretraining (CLIP) family builds a unified image-text embedding space through large-scale joint pretraining and is widely used for zero-shot and open-set visual tasks. In medical imaging, extensions such as BiomedCLIP [Zhang et al., (2023)] sharpen cross-modal understanding and perform robustly across diverse applications. Yet the capability we exploit remains untapped for our setting: existing studies apply these models mostly to downstream tasks such as object detection and semantic segmentation [Zhong et al., (2022); Aleem et al., (2024); Koleilat et al., (2024)], and none ask whether their open-vocabulary, zero-shot localization could benefit masked image modeling.

Two observations therefore define the opening our contributions fill. First, recent region-aware MIM approaches—dynamic masking from reconstruction loss, or masking guided by external perception modules—improve semantic focus only partially and remain hampered by added supervision, misalignment between masked regions and task semantics, and limited flexibility. Second, despite the strength of vision-language models downstream, almost no work injects natural-language prompts into the MIM *pretraining* phase to guide task-aware masking. Bringing vision-language semantic guidance into masking—efficiently, generalizably, and without extra supervision—is precisely the direction MWM pursues.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

This section shows how the design realizes Contribution 1 (text-guided masking) and Contribution 2 (an annotation-free, backbone-agnostic pipeline). As illustrated in Figure 1, **Mask What Matters (MWM)** comprises two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage uses a frozen vision-language model to identify task-relevant regions from an open-vocabulary prompt; the second stage applies differentiated masking ratios across those regions to steer reconstruction. Crucially, no per-image label enters either stage, which is what keeps the framework annotation-free.

3.2 TEXT-GUIDED REGION LOCALIZATION

The localization module is where Contribution 1 is instantiated: it turns an open-vocabulary prompt into a robust spatial guide without any per-image annotation. The pipeline is multi-stage. It starts with LLM-based prompt generation and uses BiomedCLIP to extract cross-modal embeddings. A saliency map is then produced via the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined with the Segment Anything Model (SAM). Finally, the refined mask be-

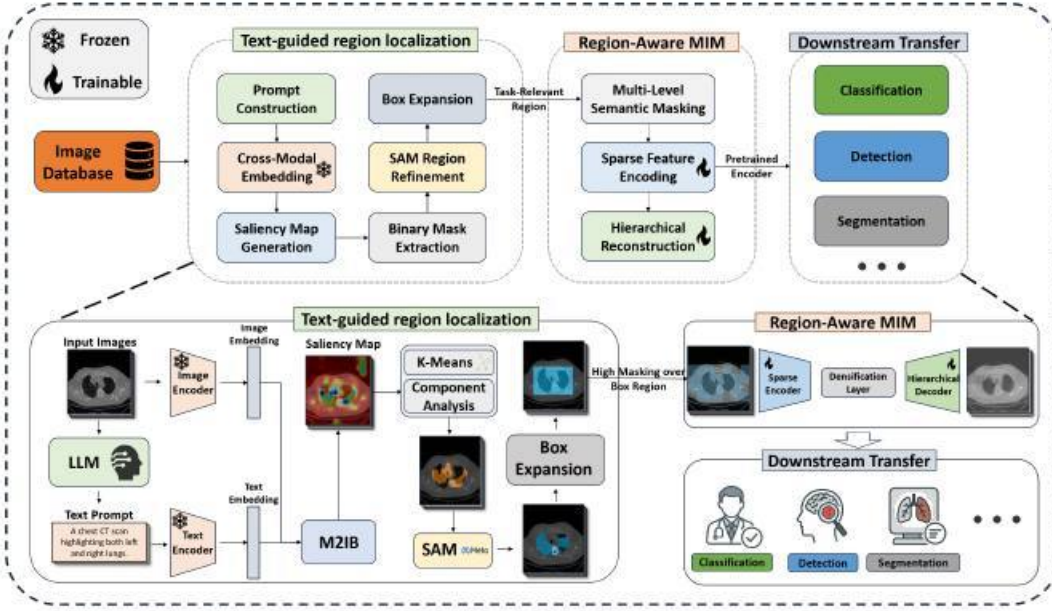


Figure 1: Overview of the MWM framework, the vehicle for all three contributions. The top panel shows the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel zooms in on how open-vocabulary prompts drive region localization and differentiated masking—the mechanism behind our annotation-free, backbone-agnostic design.

comes a bounding box with a controllable expansion margin—the robust spatial guide handed to the masked-modeling stage.

Prompt Generation. Prompt design largely determines how well region localization works in medical image analysis. To strengthen semantic guidance, large language models (LLMs) are queried with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality] image.” Alongside these full-sentence prompts, we also test simple category phrases as alternative inputs. The two prompt styles are compared in detail in the experiments, where their effect on both localization and downstream accuracy is reported.

Cross-Modal Embedding. Once the prompt is generated, a BiomedCLIP model [Zhang et al. \(2023\)](#) encodes the medical image and its prompt into a shared semantic space—the step that lets language, rather than labels, drive localization. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings are the semantic foundation on which the later saliency stages build.

Saliency Map Generation. To turn those embeddings into a spatial signal, a cross-modal saliency module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. \(2023b\)](#) localizes task-relevant regions by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ weights each spatial location by its semantic relevance to the prompt. M2IB obtains this by optimizing:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the BiomedCLIP image and text embeddings, and γ trades off preserving cross-modal relevance against filtering task-irrelevant visual information.

Binary Mask Extraction. The saliency map is then binarized with unsupervised K-Means clustering [Lloyd \(1982\)](#) to yield a preliminary, prompt-guided foreground mask. The map is split into two pixel groups, and the higher-saliency cluster is taken as foreground. Connected component analysis further cleans the mask, keeping only the largest foreground regions; this suppresses false activations and noisy edges so attention lands on the core semantic structures.

Region Refinement. For sharper spatial accuracy, SAM refines the mask [Kirillov et al. \(2023\)](#). For each selected connected region $c_i \in C^*$, its minimal enclosing bounding box becomes the visual prompt to SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} captures spatial structure more faithfully.

Box-Based Region Expansion. Because the SAM mask still carries boundary noise that can misdirect representation learning, we convert it into a bounding box with a controllable expansion margin. This box is the final region of interest (ROI) passed to masked modeling, and it makes the region guidance robust to residual boundary error—closing the localization half of Contribution 1.

3.3 REGION-AWARE MASKED IMAGE MODELING

This is where the localized ROI becomes a learning signal, realizing the masking half of Contribution 1 and the backbone-agnostic property of Contribution 2. Given the ROI produced by the localization module in Section 3.2, MWM runs a three-stage pipeline: (1) a multi-level masking strategy that assigns differentiated ratios by region importance, (2) a sparse encoder that processes only unmasked patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. This step is the mechanistic core of Contribution 1. MWM masks according to the semantic regions found by text-driven localization: patches inside prompt-highlighted regions—tumors or organs—are masked at a high ratio so the model must reconstruct task-relevant features, while background regions are masked lightly or left visible to cut redundant computation. Figure 2 contrasts this pattern with random masking.

Sparse Feature Encoding. Visible patches are gathered into a sparse image and encoded following the SparK design [Tian et al. \(2023\)](#):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are multi-scale features. We instantiate MWM with convolutional backbones here, but the same interface applies to ViTs via token indexing—this is exactly the backbone-agnostic property claimed in Contribution 2.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image top-down. At each scale l , masked locations are filled with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained with mean-squared error on masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

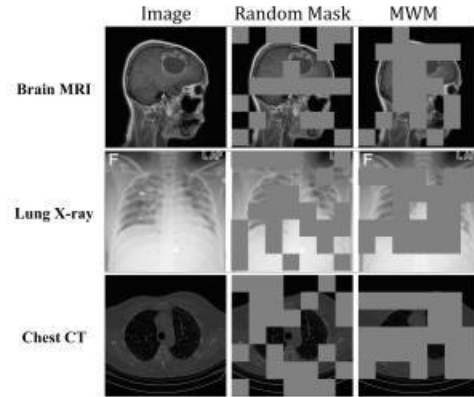


Figure 2: Comparison of masking strategies; gray blocks are masked regions. Unlike random masking, MWM concentrates masking on the prompt-localized ROI.

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction. Because the loss is computed only on masked ROI patches, the training signal is concentrated exactly where task semantics live.

4 EXPERIMENTS

Our experiments are organized to substantiate the three contributions in turn: Section 4.3 provides direct evidence for Contribution 1 (prompts localize task-relevant regions), Section 4.4 tests Contribution 3 (consistent gains across modalities and tasks), and the ablations in Section 4.5 isolate the design choices behind Contributions 1 and 2 and quantify the low-ratio advantage.

4.1 DATASETS

Self-Supervised Pretraining Datasets. To exercise the cross-modality claim in Contribution 3, we pretrain on three datasets spanning distinct imaging modalities. (1) *Brain MRI*: roughly 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering diverse tumor types and anatomical structures. (2) *Lung X-ray*: about 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021); Chowdhury et al. (2020)], spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT*: 12,000 slices from a lung disease dataset [Konya (2020)], with fibrotic-lung segmentation masks from 107 patients.

Downstream Fine-Tuning Datasets. For downstream evaluation across classification and detection, we use three further datasets. (1) *Brain Tumor MRI* [Feltrin (2023)]: 4,479 MRI images across 44 tumor types, such as astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermary et al. (2018)]: 5,863 pediatric chest X-rays labeled normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)]: CT images in four classes—adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

To keep the comparison behind Contribution 3 fair, all self-supervised methods use backbones of comparable scale: convolution-based methods use ResNet-50 [He et al. (2016)] and transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. Input resolution is 224×224 , and models start from ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} ; downstream fine-tuning keeps the same encoder and resolution. All experiments run on NVIDIA Tesla A100 cards (80 GB VRAM) under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless noted, we fix non-target variables (e.g., prompt type or SAM usage) to their empirically optimal settings, so each reported comparison isolates a single factor.

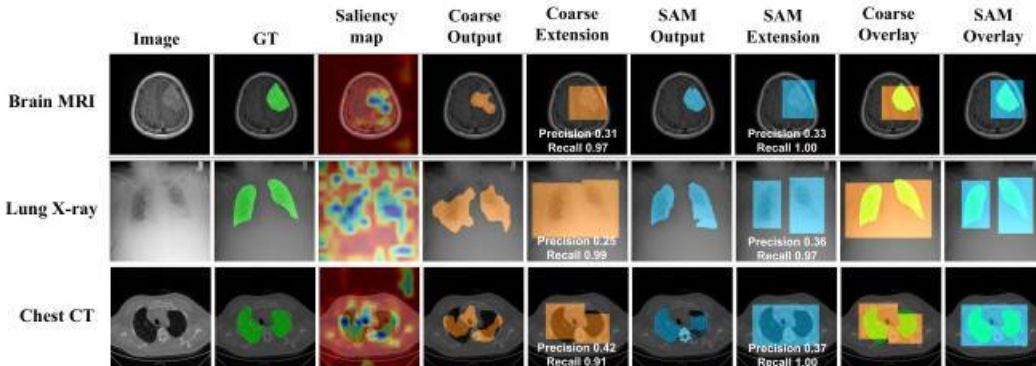


Figure 3: Representative text-guided region localization results, offering qualitative evidence for Contribution 1: prompts alone recover task-relevant regions across modalities.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Evidence for Contribution 1: localization performance (Occlusion Precision/Recall) across prompt types and refinement strategies. *Coarse Ext.* denotes bounding-box extension of the coarse region; + *SAM Ext.* adds SAM refinement. High recall across all three modalities shows prompts alone find task-relevant regions.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

This subsection delivers the first piece of evidence for Contribution 1: that open-vocabulary prompts, on their own, localize task-relevant regions. We evaluate the localization module across brain MRI, lung X-ray, and chest CT, and probe how prompt type and SAM refinement affect it. Predicted bounding boxes are compared against expert-annotated segmentation masks, reporting occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} the ground truth.

The headline result is that, with effective prompts, text-guided localization reaches consistently high occlusion recall on every dataset (around 0.82) at reasonable precision (Table 1). This confirms that MWM identifies task-relevant regions from text alone, despite modality-specific anatomical variation (Figure 3 shows examples)—the core claim of Contribution 1.

Effect of Prompt Type on Region Localization. Zooming in on language design, we compare two prompt types. Descriptive sentence prompts win on Brain MRI and Chest CT (Table 1), showing that richer semantics help localization. Concise phrases, however, give higher recall on lung radiographs (0.82 vs. 0.66), likely because the lungs’ broad anatomical coverage makes general terms sufficient. The practical takeaway is that prompt granularity should track how localized the target anatomy is.

Effect of SAM Refinement on Region Localization. Turning to spatial refinement, applying SAM to the coarse masks improves precision while holding recall high (Table 1). On Brain MRI, precision rises from 0.22 to 0.30 with recall staying above 0.94. SAM thus sharpens coarse localization without sacrificing coverage—one of the design choices we later ablate.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

This subsection is the main test of Contribution 3. We benchmark MWM’s generalization and representation quality against representative self-supervised learning (SSL) methods spanning both reconstruction-based and contrastive paradigms, pretraining and evaluating all methods under identical datasets and settings for a fair comparison. The claim to establish is a single, consistent story: MWM leads across modalities (classification), across evaluation protocols (fine-tuning and linear probing), and across task types (detection and segmentation).

Image Classification. We examine classification on the three downstream datasets under two standard protocols: (1) *full fine-tuning*, updating the whole model, and (2) *linear probing*, freezing the encoder and training only a linear classifier. Both point to the same conclusion for Contribution 3.

- **Full fine-tuning.** Table 2 reports MWM against representative self-supervised methods—MAE [He et al. (2022)], SparK [Tian et al. (2023)], AnatoMask [Li et al. (2024)], SimCLR [Chen et al. (2020a)], MoCoV2 [Chen et al. (2020b)], and BYOL [Grill et al. (2020)]. MWM is the top method on every modality, evidencing broad generalization. The largest margin is on Chest CT, where MWM reaches 97.5%—+3.1 over SparK and +1.7 over AnatoMask. The gain traces directly to text-guided masking: it concentrates on semantically critical yet spatially sparse struc-

tures (e.g., lung lobes in CT) rather than masking redundant background, which is the mechanism Contribution 1 predicts.

- **Linear probing.** To probe representation quality directly, we compare MWM with the two strongest recent MIM methods, SparK and AnatoMask. MWM outperforms both on all datasets (Table 3), confirming that text-driven pretraining learns features that are useful even without fine-tuning—a stronger form of the Contribution 3 claim.

Object Detection and Instance Segmentation. To show that Contribution 3 extends past classification, we evaluate brain tumor detection and instance segmentation on BR35H Hamada (2025). MWM attains the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), beating SparK by +1.3 and +1.1 points respectively (Table 4). MWM therefore transfers effectively to both object-level localization and pixel-wise segmentation.

Takeaway (Contribution 3). Across classification, detection, and segmentation—and under both fine-tuning and linear probing—text-driven masking during pretraining yields more informative, more transferable representations. The improvement is consistent rather than isolated, which is the crux of our third contribution.

4.5 ABLATION STUDY

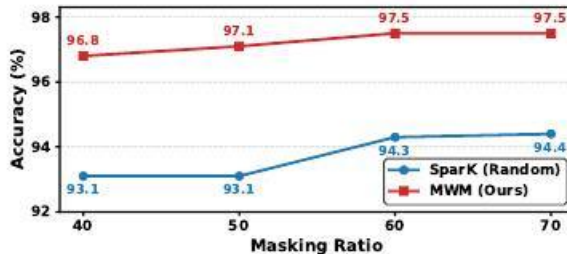
The ablations tie each empirical gain back to a specific design choice, so the contributions rest on identified causes rather than aggregate numbers. We isolate the effect of prompt design, SAM-based refinement, and masking ratio on downstream classification.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

Table 5 makes two points central to Contribution 1. First, descriptive sentence prompts beat short phrases on brain MRI and chest CT while the reverse holds on lung X-ray—mirroring the localization results of Section 4.3, so better region identification reliably yields better downstream accuracy. Second, and most importantly, *any* text prompt, regardless of granularity, beats random masking on every dataset. This is direct evidence that text-guided masking aligns pretraining with downstream semantics, the mechanism our first contribution claims. The SAM ablation adds a smaller but consistent improvement, confirming that region refinement further sharpens text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

This experiment substantiates the efficiency half of Contribution 3—that MWM’s gains come *with* lower masking ratios, not despite them. MWM reaches 96.8% accuracy at a masking ratio of just 40% (Figure 4), far below the 70% typical of masked image modeling, and still beats SparK’s best 70% result by +2.4% abso-



Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	91.3	92.7	93.9
BYOL	CL	91.3	92.7	93.9
SimCLR	CL	91.3	92.7	93.9
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Contribution 3, fine-tuning view: classification accuracy (%) on downstream datasets (MIM = Masked Image Modeling, CL = Contrastive Learning). MWM (shaded) is best on all three modalities.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Contribution 3, representation-quality view: linear-probing accuracy (%) with a frozen encoder. MWM (shaded) leads on all datasets.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Contribution 3 beyond classification: detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) (%) on BR35H. MWM (shaded) leads on detection and is competitive on segmentation.

lute. To be sure SparK is not merely mistuned, we sweep it over 40%, 50%, 60%, and 70%; its accuracy peaks at 70%, so the comparison is fair. The upshot challenges the prevailing assumption that high masking ratios are inherently optimal: semantically guided, task-aware masking preserves task-relevant content and drops redundancy, delivering stronger downstream performance at lower ratios.

5 CONCLUSION AND LIMITATIONS

We presented **Mask What Matters (MWM)**, a self-supervised pretraining framework that unites text-guided semantic localization with region-aware masking. Using natural-language prompts to highlight task-relevant areas and applying differentiated masking, MWM learns semantically aligned representations without any per-image annotation. Restating the three contributions in light of the evidence:

- **Contribution 1 (text-guided masking).** Natural-language prompts reliably localize semantic regions (Section 4.3, Table 1), and text-guided masking beats random masking on every dataset (Table 5)—validating the mechanism, not just the outcome.
- **Contribution 2 (annotation-free, backbone-agnostic design).** The whole pipeline runs without per-image labels and plugs into ConvNet- and ViT-based MIM alike under one protocol, making it broadly reusable.
- **Contribution 3 (consistent gains at lower cost).** MWM outperforms existing methods (e.g., SparK, AnatoMask) in classification across three modalities and improves detection and instance segmentation, all while remaining strong at a masking ratio as low as 40%.

Limitations. Two limitations temper these contributions. First, although prompts give flexible guidance, we have not systematically tested MWM’s robustness to variation in prompt style, vocabulary, or noise. Second, the dependence on external textual descriptions is itself a risk: generalization may weaken wherever reliable prompts are unavailable or inconsistent.

Looking forward, text-driven, region-aware masking is, we argue, a principled route to more semantically grounded self-supervised learning, and carrying MWM beyond medical imaging may open broader opportunities for vision–language pretraining.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Ablation isolating the sources of Contributions 1 and 2: downstream classification accuracy (%) under different prompt types and refinement strategies across three modalities. *No Prompt* is random masking without text-guided localization; *w/* and *w/o* denote with and without SAM. Any text prompt beats *No Prompt*, and SAM adds a further consistent gain.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

We used LLMs *only* to edit language and polish presentation (grammar, phrasing, and minor stylistic clarity) of text the authors wrote. We did *not* use LLMs for idea generation, experimental design, data annotation, code implementation, or analysis and interpretation of results. All technical content—algorithms, proofs, figures, tables, metrics, and conclusions—was authored and verified by the authors, and no confidential or identifying data beyond anonymized manuscript text was ever provided to an LLM.